ECE719: Advanced Microwave Design
Instructor: Prof. Suresh Venkatesh, NCSU.

Project 2: Holographic Beamforming

First, we will setup a basic tunable radiator and then build a linear array using these tunable radiators followed by beamforming analysis.

Response function:

Let's start with modelling a radiator as a Lorentzian. The amplitude and phase profile of the Lorentzian as follows:

Following functions are trying to approximate the responses shown in the article (**Reconfigurable Reflectarrays and Array Lenses for Dynamic Antenna Beam Control: A Review**, figure. 9). Amplitude and phase responses are coupled.

Approximations of this response can be done as follows:

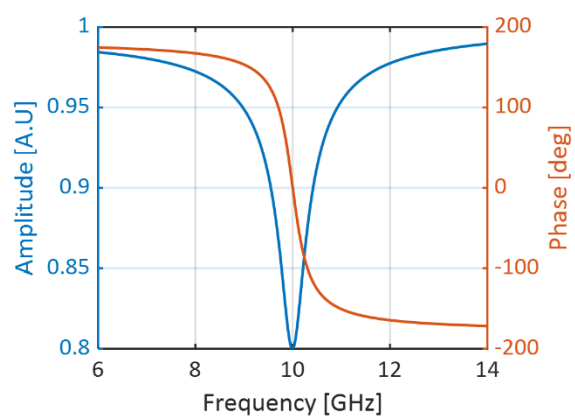
$$R = \frac{2+f_r}{f_r^2 - f^2 + i\,\Delta f * f}$$

Amplitude response of the radiator = $1 - \{abs(R) * 100e6\}$

Phase response of the radiator = $[2 * angle(R) + \Pi]$ (in radians)

f_r *is the resonant frequency*
Δf is the damping factor (assume 0.05 f_r)
f *is frequency*

f_r = 10 GHz, Δf = 0.05 f_r and f from 6 GHz to 14 GHz, the amplitude (blue)and phase response (red in degrees) is shown in the right figure.



Tunability of the Lorentzian:

Now that we have defined the response of a radiator, let's say we can tune this Lorentzian response through a varactor (ideal case, no loss). Mathematically, it is just changing the resonant frequency f_r

For example, if the resonant frequency is changed to 12 GHz, the response changes as shown in the dotted line. (The resonant frequency can along change from 8 to 12 GHz (X-band))

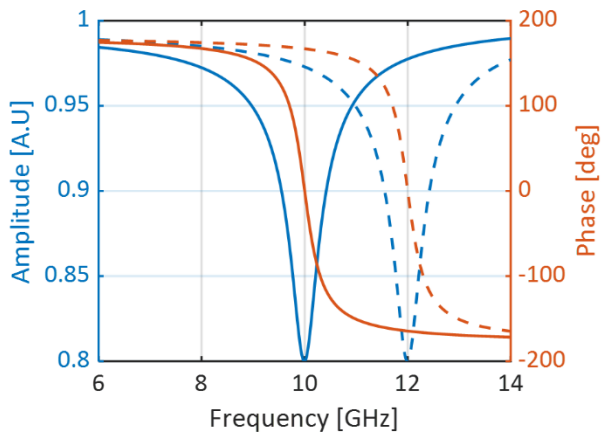


Figure 1: Resonant Frequency fr at 12 GHz

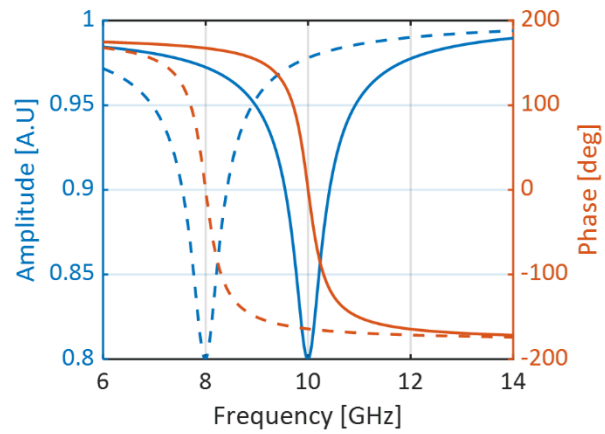


Figure2: Resonant Frequency fr at 8 GHz

Phase shifter:

Now the tunable Lorentzian is our phase shifter achieved by changing the resonant frequency from 8 to 12 GHz. (Note the corresponding the amplitude could also change, which is a drawback).

As an example, if we have 2 bit phase shifter at 10 GHz we can think about tuning the Lorentzian from 8 to 12 GHz as follows.

To get a particular phase, we can now solve for the resonant frequency based on equations we have. In MATLAB, you can do it as follows: This will give the resonant frequency that achieves a phase shift of -90 degrees at 10 GHz.

```
f = 10e9;
syms fr;
eqn = ((2*angle((2 + fr)./(fr^2-f^2+1i*0.05*fr*f))+pi)*180/pi ) == -90;
assume(fr,'positive')
S = (vpasolve(eqn,fr))
```

Now we know how to tune our resonant frequency to achieve a particular phase shift at the center frequency of 10 GHz. **Remember we will only use available phases that have a resonant frequency from 8 to 12 GHz (X-band).**

Based on the resonant frequency you can now find the associated amplitude that goes along with the phase shift.

As an example, a 2-bit Phase shifter at 10 GHz can be achieved by tuning the resonator at 4 different resonant frequencies as shown below:

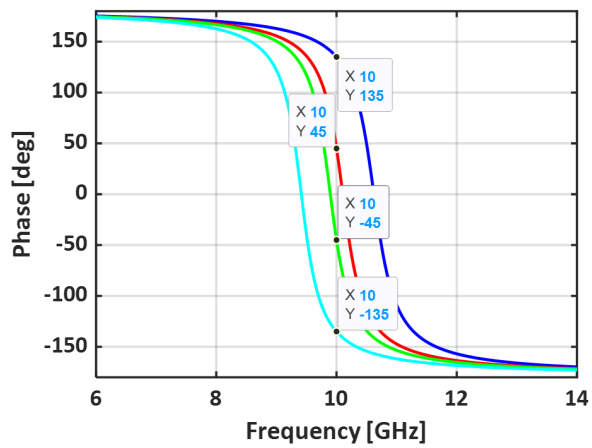


Figure 3: Shows the phases achievable after quantization

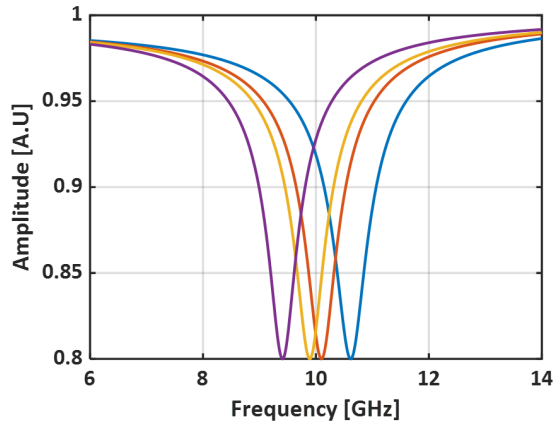


Figure 4: Shows the corresponding amplitude for the phases

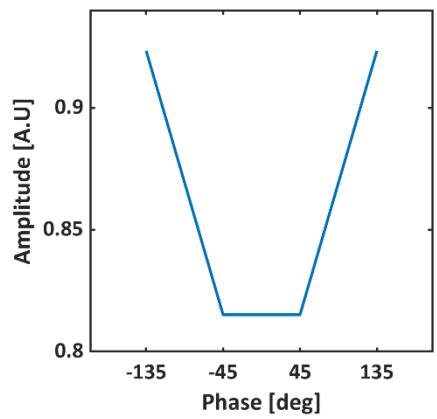


Figure 5: amplitude versus phase @ 10 GHz

Till here is the basic setup of the phase shifter element (that has couple amplitude and phase responses)

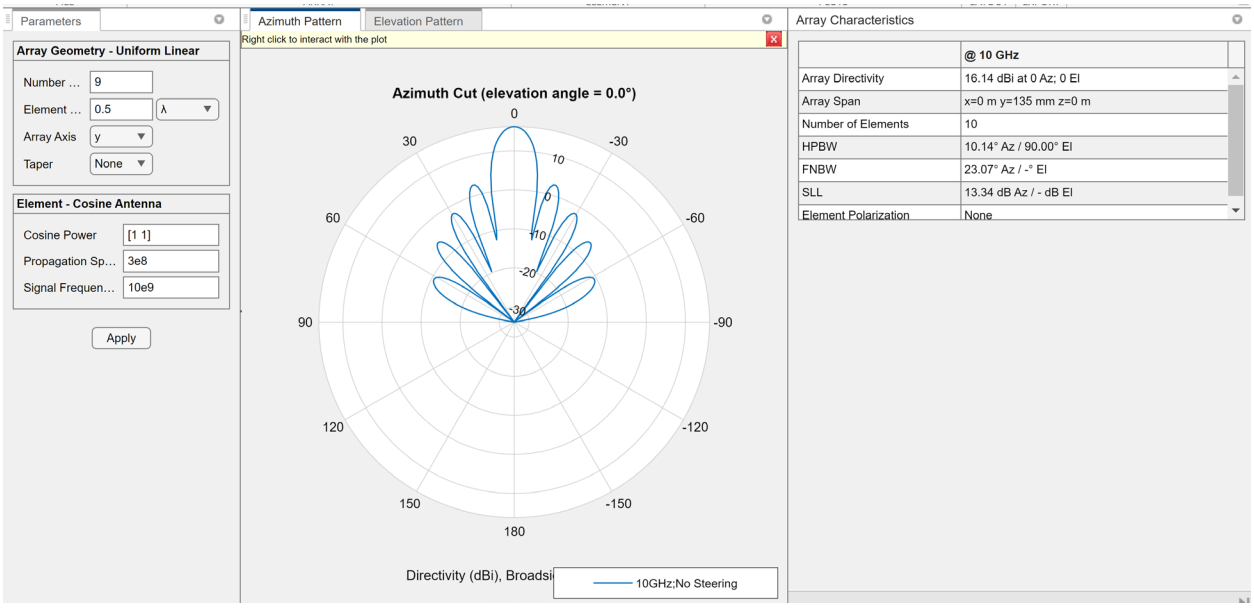
For Python users: Use the equivalent equation solver function.

RF Beamforming Setup:

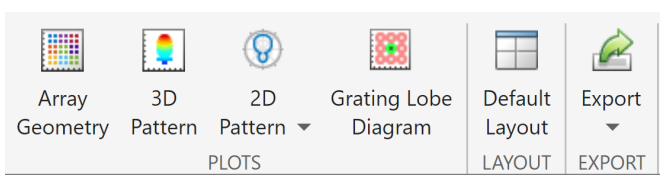
So this will be our basic setup to do beamforming. Using this we will perform beamforming and look at different aspects of the array.

We will use MATLAB phased array toolbox to perform broadside beamforming and derive some of the metrics. Type 'sensorArrayAnalyzer' to get the GUI below on MATLAB

- **Number of resonating radiators = 9 Uniform Linear Array**
- **Center Frequency = 10 GHz**
- **At each element we assume cosine pattern.**
- **Spacing between the elements is Lambda/2 = 3 cm**
- **Antenna element – choose cosine element with cosine powers [2 2]**



Use Export and Generate Matlab Script. Use this script now to enter your phase matrix and amplitude matrix corresponding to each element.



In the generated Script replace the weights with w with your own weights you determine from the Lorentzian function (both amplitude and phase and convert to a complex form x+iy)

```
% Find the weights
w = ones(getNumElements(Array), length(Frequency));
```

Note: Amplitude will be determined from the phase shift you would need to apply to that element. Phase shift will be determined from the pointing angle you need to achieve. Not all phase shifts can be achieved depending on the phase quantization and you will need to find the nearest value.

```
% Create a cosine antenna element
Elem = phased.CosineAntennaElement;
Elem.CosinePower = [1.8 1.8];
Elem.FrequencyRange = [0 10e9];
Array.Element = Elem;
% Create Figure
```

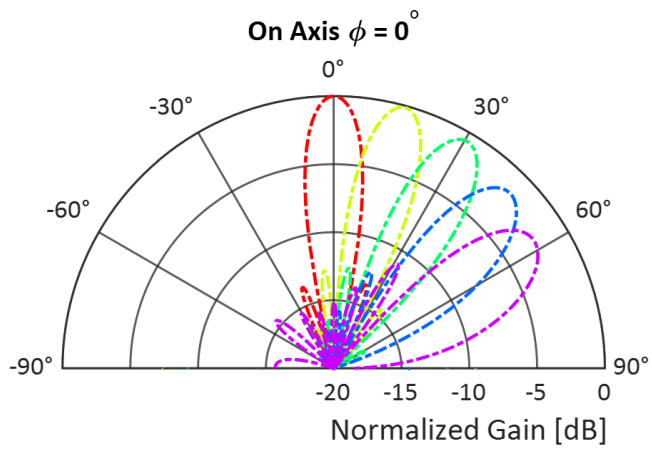
Choose the following properties for the antenna element. Use the powers [2 2] to ensure some exaggerated roll-off.

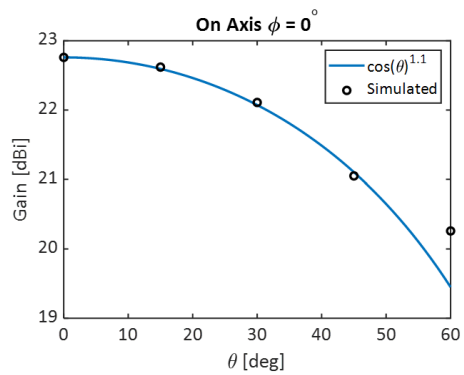
For Python Users: You will have to write a code that calculates the array factor and multiply with the element pattern for different cases.

ECE719: Advanced Microwave Design
Instructor: Prof. Suresh Venkatesh, NCSU.

Problem1: Assume you have continuous resonant tuning from 8 to 12 GHz only.
a) Plot the intended pointing angle versus error in the pointing angle at 10 GHz.
(error = abs(intended point angle – actual pointing angle)).
Vary the pointing angle from 0 to 60 degrees in steps of 10 or 15 degrees.

b) Plot the scan roll-off for the continuous tuning case (function of intended pointing angle and maximum gain achieved at that intended pointing angle). Normalize your beam plots to 0 degree pointing angle. Fit to a curve of the form $(\cos(\theta))^n$, where θ is the intended pointing angle and n is a constant around 1.1 – 1.4. Show the equation of your fitted curve. Here is an example:





Problem 2: Repeat problem 1 case a) and b) for different quantization and compare with problem 1.
c) Quantization: 2 bits with 4 phase states
d) Quantization: 3 bits with 8 phase states
e) Quantization: 4 bits with 16 phase states
Note: In each of the quantization states show the amplitude and phase plots of the Lorentzian you have chosen. (as shown in Figure 5, extracted from figure 3 and figure 4)

Also comment on the following:
f) Comment and show the minimum number of quantization levels required to achieve the continuous case (problem 1)

g) Show the effect of side lobe levels as a function of quantization and pointing angles.

Project Report:

1. Provide a brief executive summary of the project.
2. Clearly show the plots of the beamforming and associated plots with proper labels and units along the axis.
3. Summarize your observation for each problem and relate it to the theoretical value.
4. Good polar plots showing steering clearly with roll-off and sibe-lobe effects will be considered for **extra credits** (see the sample plots above). Just snapping the Matlab plot won't be sufficient.